Ethical Hacking Roadmap

A complete, step-by-step learning path designed for beginners to master cybersecurity fundamentals, network security, system defence, and hands-on ethical hacking methodology.

Phase 1: Computer Fundamentals

Core Basics

- Computer Hardware Fundamentals
- Operating Systems Fundamentals
- Windows Architecture
- Linux Architecture
- Windows Command Line
- Linux Terminal Commands
- File Systems
- Processes and Services
- User Accounts and Permissions
- Virtualization Fundamentals

Phase 2: Networking Fundamentals

Core Basics

- Networking Basics & Topologies
- OSI Model & TCP/IP Model
- IP Addressing (IPv4 & IPv6) & Subnetting
- MAC Addressing & ARP Protocol
- TCP, UDP & ICMP Protocols
- DNS, DHCP & NAT Protocols
- Routing, Switching & VLANs
- Common Ports & Services
- Firewalls, VPN & Proxy Servers
- Wireless Networking Fundamentals

Phase 3: Linux for Ethical Hacking

OS Mastery

- Kali Linux Fundamentals
- Linux File System & Permissions
- User and Group Management
- Process & Service Management
- Package Management
- SSH Configuration & Usage
- System Logs Analysis
- Linux Security Fundamentals

Phase 4: Reconnaissance & OSINT

Recon

- Passive & Active Reconnaissance
- OSINT Fundamentals & WHOIS Lookup
- DNS Enumeration & IP Discovery
- Network & Host Discovery
- Technology Identification & Footprinting
- Google Dorking Techniques
- Social Media & Email Intelligence
- Metadata Analysis

Phase 5: Scanning & Enumeration

Enumeration

- Network & Port Scanning
- Service & OS Detection
- Vulnerability Scanning & Mapping
- Banner Grabbing Techniques
- SMB & SNMP Enumeration
- NetBIOS & LDAP Enumeration
- FTP, SSH & NFS Enumeration

Phase 6: System Hacking

Exploitation

- Authentication Fundamentals
- Password Security & Cracking Concepts
- Password Hashing & Hash Identification
- Windows & Linux Authentication
- Privilege Escalation (Windows & Linux)
- Persistence Mechanisms & Access Control
- System Hardening & Post-Exploitation

Phase 7: Wireless Network Security

Wireless

- Wireless Architecture & 802.11 Standards
- Wi-Fi Authentication & Encryption
- WEP, WPA, WPA2 & WPA3 Security
- Wireless Reconnaissance & Discovery
- Packet & Handshake Analysis
- Wireless Security Auditing
- Rogue AP & Evil Twin Attack Concepts
- Wireless Network Hardening

Phase 8: Network Hacking & Security

Network

- Network Traffic Analysis & Packet Sniffing
- Protocol Analysis & ARP/MAC Spoofing
- DNS Spoofing & MITM Attack Concepts
- Session Hijacking Concepts
- DoS and DDoS Concepts
- Network Segmentation & Firewall Evasion
- Network Auditing & Hardening

Phase 9: Social Engineering

Human Vector

- Social Engineering Fundamentals
- Human-Based vs Computer-Based Attacks
- Phishing, Spear Phishing & Whaling
- Vishing, Smishing & Pretexting
- Baiting, Tailgating & Impersonation
- USB-Based Social Engineering
- Awareness & Prevention Strategies

Phase 10: Mobile Security

Mobile

- Android Architecture & Security Model
- Android Permissions & App Security
- Android Debug Bridge (ADB) Fundamentals
- Android Device Forensics
- Mobile Network Security & Authentication
- iOS Security Architecture
- Mobile Device Hardening & MDM

Phase 11: Cryptography

Crypto

- Cryptography Fundamentals
- Symmetric & Asymmetric Encryption
- Hashing Algorithms & Digital Signatures
- Public Key Infrastructure (PKI)
- SSL/TLS Fundamentals & Certificates
- Key Exchange & Encryption Standards
- Cryptographic Attack Concepts

Phase 12: Vulnerability Assessment

Assessment

- Vulnerability Identification & Classification
- CVE, CVSS, and CWE Frameworks
- Vulnerability Scanning & Risk Assessment
- False Positive Identification
- Vulnerability Prioritization & Remediation
- Security Patch Management

Phase 13: System & Network Defense

Blue Team

- Endpoint Security, Antivirus & EDR
- Intrusion Detection & Prevention (IDS/IPS)
- Firewall Configuration & SIEM Fundamentals
- Log Analysis & Security Monitoring
- Incident Response & Threat Intelligence
- Digital Forensics & Malware Analysis Basics
- Security Auditing & Disaster Recovery

Phase 14: Practical Labs & Skills

Hands-On

- Kali Linux, Windows & Linux Security Labs
- Network Scanning & Packet Analysis Labs
- Vulnerability Assessment & Password Labs
- Wireless Security & System Hardening Labs
- Mobile Security Lab & Network Audits
- Penetration Testing Methodology
- Rules of Engagement & Legal Guidelines
- Penetration Testing Report Writing